KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3241 | 3 Credits: 3 | Engineering Science | Source-grounded

Basic Medical Sciences

 To lay foundation about the basic medical terms used in biomedical engineering and

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3241 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3241.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Biomedical Engineering
Course code	3241
Course title	Basic Medical Sciences
Semester	3 Credits: 3
Course category	Engineering Science
Periods per week	3 (L:2, T:1, P:0) Periods per semester: 45

Item	Official information
Periods per semester	45
Credits	3
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To lay foundation about the basic medical terms used in biomedical engineering and
- basic anatomy and physiology of the human body.
- To provide the knowledge of anatomy and physiology for devising technology for
- human rehabilitation.

Course outcomes

CO	Official outcome	Study level
CO1	12 Understanding transmission in human body. Describe the anatomy and physiology of the	See official syllabus
CO2	11 Understanding cardiovascular and respiratory systems.	See official syllabus
CO3	Illustrate the Nervous System and Urinary System 11 Understanding Explain the anatomy of the eye and ear and	See official syllabus
CO4	9 Understanding functioning of all special sense organs	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Summarize bio-potential generation and transmission in human body.	4	As specified
Module 2	respiratory systems.	5	As specified
Module 3	Illustrate the Nervous System and Urinary System	4	As specified
Module 4	organs	3	As specified

MODULE 1

Summarize bio-potential generation and transmission in human body.

This module is presented from the official syllabus scope for Basic Medical Sciences. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize bio-potential generation and transmission in human body.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Define various medical terminologies 2 Remembering Describe the anatomy and physiology of cell and

Define various medical terminologies 2 Remembering Describe the anatomy and physiology of cell and is an official topic in Module 1 of Basic Medical Sciences. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define various medical terminologies 2 Remembering Describe the anatomy and physiology of cell and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define various medical terminologies 2 remembering describe the anatomy and physiology of cell and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define various medical terminologies 2 Remembering Describe the anatomy and physiology of cell and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Define various medical terminologies 2 Remembering Describe the anatomy and physiology of cell and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

3 Understanding tissues Illustrate the concept of bioelectric potentials

3 Understanding tissues Illustrate the concept of bioelectric potentials is an official topic in Module 1 of Basic Medical Sciences. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding tissues Illustrate the concept of bioelectric potentials using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding tissues illustrate the concept of bioelectric potentials should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding tissues Illustrate the concept of bioelectric potentials means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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but different term.

Malayalam support: Module 1-□□ 3 Understanding tissues Illustrate the concept of bioelectric potentials □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

3 Understanding and demonstrate using Nernst Equation. Extend the concept of bio potential

3 Understanding and demonstrate using Nernst Equation. Extend the concept of bio potential is an official topic in Module 1 of Basic Medical Sciences. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding and demonstrate using Nernst Equation. Extend the concept of bio potential using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding and demonstrate using nernst equation. extend the concept of bio potential should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding and demonstrate using Nernst Equation. Extend the concept of bio potential means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-▯▯ 3 Understanding and demonstrate using Nernst Equation. Extend the concept of bio potential ▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ definition/meaning ▯▯▯▯▯▯▯▯▯▯▯▯▯▯. ▯▯▯▯▯▯ ▯▯▯▯▯▯ ▯▯▯▯▯▯▯, steps, application ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯. Technical terms English-▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯.

3 Understanding transmission in nerves and its measurement. Cell, Tissues and Bio potential Definition of anatomy, physiology, biochemistry, biophysics. Basic anatomical positions and planes - supine, prone, sagittal, coronal and transverse. Cells: structure of human cell, Elementary tissues of the body - types and their functions(listing) - Epithelial, connective, muscle, nerve cells Introduction to bio potential, Origin of bioelectric potential, Resting membrane potential, Action potential, Transmission of impulse through nerve fibers, Recording of membrane potential - microelectrode, monophasic, biphasic. Describe the anatomy and physiology of the cardiovascular and

3 Understanding transmission in nerves and its measurement. Cell, Tissues and Bio potential Definition of anatomy, physiology, biochemistry, biophysics. Basic anatomical positions and planes - supine, prone, sagittal, coronal and transverse. Cells: structure of human cell, Elementary tissues of the body - types and their functions(listing) - Epithelial, connective, muscle, nerve cells Introduction to bio potential, Origin of bioelectric potential, Resting membrane potential, Action potential, Transmission of impulse through nerve fibers, Recording of membrane potential - microelectrode, monophasic, biphasic. Describe the anatomy and physiology of the cardiovascular and is an official topic in Module 1 of Basic Medical Sciences. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding transmission in nerves and its measurement. Cell, Tissues and Bio potential Definition of anatomy, physiology, biochemistry, biophysics. Basic anatomical positions and planes - supine, prone, sagittal, coronal and transverse. Cells: structure of human cell, Elementary tissues of the body - types and their functions(listing) - Epithelial, connective, muscle, nerve cells Introduction to bio potential, Origin of bioelectric potential, Resting membrane potential, Action potential, Transmission of impulse through nerve fibers, Recording of membrane potential - microelectrode, monophasic, biphasic. Describe the anatomy and physiology of the cardiovascular and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding transmission in nerves and its measurement. cell, tissues and bio potential definition of anatomy, physiology, biochemistry, biophysics. basic anatomical positions and planes - supine, prone, sagittal, coronal and transverse. cells: structure of human cell, elementary tissues of the body - types and their functions(listing) - epithelial, connective, muscle, nerve cells introduction to bio potential, origin of bioelectric potential, resting membrane potential, action potential, transmission of impulse through nerve fibers, recording of membrane potential - microelectrode, monophasic, biphasic. describe the anatomy and physiology of the cardiovascular and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding transmission in nerves and its measurement. Cell, Tissues and Bio potential Definition of anatomy, physiology, biochemistry, biophysics. Basic anatomical positions and planes - supine, prone, sagittal, coronal and transverse. Cells: structure of human cell, Elementary tissues of the body - types and their functions(listing) - Epithelial, connective, muscle, nerve cells Introduction to bio potential, Origin of bioelectric potential, Resting membrane potential, Action potential, Transmission of impulse through nerve fibers, Recording of membrane potential - microelectrode, monophasic, biphasic. Describe the anatomy and physiology of the cardiovascular and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 3 Understanding transmission in nerves and its measurement. Cell, Tissues and Bio potential Definition of anatomy, physiology, biochemistry, biophysics. Basic anatomical positions and planes - supine, prone, sagittal, coronal and transverse. Cells: structure of human cell, Elementary tissues of the body - types and their functions(listing) - Epithelial, connective, muscle, nerve cells Introduction to bio potential, Origin of bioelectric potential, Resting membrane potential, Action potential, Transmission of impulse through nerve fibers, Recording of membrane potential - microelectrode, monophasic, biphasic. Describe the anatomy and physiology of the cardiovascular and □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

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Summarize bio-potential generation and transmission in human body.

  M1.01      Define various medical terminologies                        2
Remembering
             Describe the anatomy and physiology of cell and
  M1.02                                                                  3
Understanding
             tissues
             Illustrate the concept of bioelectric potentials
  M1.03                                                                  3
Understanding
             and demonstrate using Nernst Equation.
             Extend the concept of bio potential
  M1.04                                                                  3
Understanding
             transmission in nerves and its measurement.

Contents:
Cell, Tissues and Bio potential
Definition of anatomy, physiology, biochemistry, biophysics. Basic anatomical
positions
and planes - supine, prone, sagittal, coronal and transverse.
Cells: structure of human cell, Elementary tissues of the body - types and their
functions(listing) - Epithelial, connective, muscle, nerve cells
Introduction to bio potential, Origin of bioelectric potential, Resting membrane
potential,
Action potential, Transmission of impulse through nerve fibers, Recording of
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

respiratory systems.

This module is presented from the official syllabus scope for Basic Medical Sciences. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: respiratory systems.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

2 Understanding circulation of blood in human body Summarize the anatomy and physiology of

2 Understanding circulation of blood in human body Summarize the anatomy and physiology of is an official topic in Module 2 of Basic Medical Sciences. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding circulation of blood in human body Summarize the anatomy and physiology of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding circulation of blood in human body summarize the anatomy and physiology of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding circulation of blood in human body Summarize the anatomy and physiology of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 2 Understanding circulation of blood in human body Summarize the anatomy and physiology of □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

2 Understanding human heart Illustrate the electrical activity of the heart and

2 Understanding human heart Illustrate the electrical activity of the heart and is an official topic in Module 2 of Basic Medical Sciences. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding human heart Illustrate the electrical activity of the heart and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding human heart illustrate the electrical activity of the heart and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding human heart Illustrate the electrical activity of the heart and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-[illegible] 2 Understanding human heart Illustrate the electrical activity of the heart and [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Understanding cardiac cycle Describe the anatomy and physiology of the

3 Understanding cardiac cycle Describe the anatomy and physiology of the is an official topic in Module 2 of Basic Medical Sciences. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding cardiac cycle Describe the anatomy and physiology of the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding cardiac cycle describe the anatomy and physiology of the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding cardiac cycle Describe the anatomy and physiology of the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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respiratory system & physics involved in 2 Understanding mechanism of respiration Explain the mechanism of ventilation, lung

respiratory system & physics involved in 2 Understanding mechanism of respiration Explain the mechanism of ventilation, lung is an official topic in Module 2 of Basic Medical Sciences. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify respiratory system & physics involved in 2 Understanding mechanism of respiration Explain the mechanism of ventilation, lung using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, respiratory system & physics involved in 2 understanding mechanism of respiration explain the mechanism of ventilation, lung should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What respiratory system & physics involved in 2 Understanding mechanism of respiration Explain the mechanism of ventilation, lung means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ respiratory system & physics involved in 2 Understanding mechanism of respiration Explain the mechanism of ventilation, lung □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

2 Understanding volumes and lung capacities Cardiovascular and Respiratory Systems The cardiovascular system - systemic and pulmonary circulation, The structure of heart - chambers and valves - major blood vessels, The functioning of heart - electrical and mechanical events - cardiac cycle - blood pressure - cardiac output, ECG Respiratory System: Anatomy of the respiratory system, Mechanism of ventilation, Process of gas exchange, Definition of lung volumes and capacities.

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Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding volumes and lung capacities Cardiovascular and Respiratory Systems The cardiovascular system - systemic and pulmonary circulation, The structure of heart - chambers and valves - major blood vessels, The functioning of heart - electrical and mechanical events - cardiac cycle - blood pressure - cardiac output, ECG Respiratory System: Anatomy of the respiratory system, Mechanism of ventilation, Process of gas exchange, Definition of lung volumes and capacities. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding volumes and lung capacities cardiovascular and respiratory systems the cardiovascular system - systemic and pulmonary circulation, the structure of heart - chambers and valves - major blood vessels, the functioning of heart - electrical and mechanical events - cardiac cycle - blood pressure - cardiac output, ecg respiratory system: anatomy of the respiratory system, mechanism of ventilation, process of gas exchange, definition of lung volumes and capacities. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding volumes and lung capacities Cardiovascular and Respiratory Systems The cardiovascular system - systemic and pulmonary circulation, The structure of heart - chambers and valves - major blood vessels, The functioning of heart - electrical and mechanical events - cardiac cycle - blood pressure - cardiac output, ECG Respiratory System: Anatomy of the respiratory system, Mechanism of ventilation, Process of gas exchange, Definition of lung volumes and capacities. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 2 Understanding volumes and lung capacities Cardiovascular and Respiratory Systems The cardiovascular system - systemic and pulmonary circulation, The structure of heart - chambers and valves - major blood vessels, The functioning of heart - electrical and mechanical events - cardiac cycle - blood pressure - cardiac output, ECG Respiratory System: Anatomy of the respiratory system, Mechanism of ventilation, Process of gas exchange, Definition of lung volumes and capacities. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
respiratory systems.
             Explain the cardiovascular system and the
  M2.01                                                                        2
Understanding
             circulation of blood in human body
             Summarize the anatomy and physiology of
  M2.02                                                                        2
Understanding
             human heart
             Illustrate the electrical activity of the heart and
  M2.03                                                                        3
Understanding
             cardiac cycle
            Describe the anatomy and physiology of the
  M2.04     respiratory system & physics involved in                    2
Understanding
            mechanism of respiration
            Explain the mechanism of ventilation, lung
  M2.05                                                                 2
Understanding
            volumes and lung capacities
            Series Test – I                                             1

Contents:
Cardiovascular and Respiratory Systems
The cardiovascular system - systemic and pulmonary circulation, The structure of
heart -
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Illustrate the Nervous System and Urinary System

This module is presented from the official syllabus scope for Basic Medical Sciences. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Illustrate the Nervous System and Urinary System
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

4 Understanding system and anatomy of brain Summarize the physiology of each part of the

4 Understanding system and anatomy of brain Summarize the physiology of each part of the is an official topic in Module 3 of Basic Medical Sciences. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding system and anatomy of brain Summarize the physiology of each part of the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding system and anatomy of brain summarize the physiology of each part of the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding system and anatomy of brain Summarize the physiology of each part of the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 4 Understanding system and anatomy of brain Summarize the physiology of each part of the □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Understanding brain Summarize the anatomy and physiology of the

2 Understanding brain Summarize the anatomy and physiology of the is an official topic in Module 3 of Basic Medical Sciences. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding brain Summarize the anatomy and physiology of the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding brain summarize the anatomy and physiology of the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding brain Summarize the anatomy and physiology of the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-□□ 2 Understanding brain Summarize the anatomy and physiology of the □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

2 Understanding urinary system Illustrate the structure of kidneys, nephron and

2 Understanding urinary system Illustrate the structure of kidneys, nephron and is an official topic in Module 3 of Basic Medical Sciences. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding urinary system Illustrate the structure of kidneys, nephron and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding urinary system illustrate the structure of kidneys, nephron and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding urinary system Illustrate the structure of kidneys, nephron and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 2 Understanding urinary system Illustrate the structure of kidneys, nephron and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

3 Understanding explain the process of urine formation Nervous system and Urinary Systems General organization of the nervous system, Central nervous system - anatomy and functioning of the human brain - cerebrum - cerebellum - brainstem - pons - hypothalamus(temperature regulation) - medulla, Spinal cord Urinary System: anatomy of urinary system - overview of anatomy of the kidney, Structure and function of the nephron, physiology of urine formation Explain the anatomy of the eye and ear and functioning of all special sense

3 Understanding explain the process of urine formation Nervous system and Urinary Systems General organization of the nervous system, Central nervous system - anatomy and functioning of the human brain - cerebrum - cerebellum - brainstem - pons - hypothalamus(temperature regulation) - medulla, Spinal cord Urinary System: anatomy of urinary system - overview of anatomy of the kidney, Structure and function of the nephron, physiology of urine formation Explain the anatomy of the eye and ear and functioning of all special sense is an official topic in Module 3 of Basic Medical Sciences. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding explain the process of urine formation Nervous system and Urinary Systems General organization of the nervous system, Central nervous system - anatomy and functioning of the human brain - cerebrum - cerebellum - brainstem - pons - hypothalamus(temperature regulation) - medulla, Spinal cord Urinary System: anatomy of urinary system - overview of anatomy of the kidney, Structure and function of the nephron, physiology of urine formation Explain the anatomy of the eye and ear and functioning of all special sense using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding explain the process of urine formation nervous system and urinary systems general organization of the nervous system, central nervous system - anatomy and functioning of the human brain - cerebrum - cerebellum - brainstem - pons - hypothalamus(temperature regulation) - medulla, spinal cord urinary system: anatomy of urinary system - overview of anatomy of the kidney, structure and function of the nephron, physiology of urine formation explain the anatomy of the eye and ear and functioning of all special sense should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding explain the process of urine formation Nervous system and Urinary Systems General organization of the nervous system, Central nervous system - anatomy and functioning of the human brain - cerebrum - cerebellum - brainstem - pons - hypothalamus(temperature regulation) - medulla, Spinal cord Urinary System: anatomy of urinary system - overview of anatomy of the kidney, Structure and function of the nephron, physiology of urine formation Explain the anatomy of the eye and ear and functioning of all special sense means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 3 Understanding explain the process of urine formation Nervous system and Urinary Systems General organization of the nervous system, Central nervous system - anatomy and functioning of the human brain - cerebrum - cerebellum - brainstem - pons - hypothalamus(temperature regulation) - medulla, Spinal cord Urinary System: anatomy of urinary system - overview of anatomy of the kidney, Structure and function of the nephron, physiology of urine formation Explain the anatomy of the eye and ear and functioning of all special sense □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Illustrate the Nervous System and Urinary System
            Explain the general organization of the nervous
  M3.01                                                       4
Understanding
            system and anatomy of brain
            Summarize the physiology of each part of the
  M3.02                                                       2
Understanding
            brain
            Summarize the anatomy and physiology of the
  M3.03                                                       2
Understanding
            urinary system
            Illustrate the structure of kidneys, nephron and
  M3.04                                                       3
Understanding
            explain the process of urine formation

Contents:
Nervous system and Urinary Systems
General organization of the nervous system, Central nervous system - anatomy and
functioning of the human brain - cerebrum - cerebellum - brainstem - pons -
hypothalamus(temperature regulation) - medulla, Spinal cord
Urinary System: anatomy of urinary system - overview of anatomy of the kidney,
Structure
and function of the nephron, physiology of urine formation
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

organs

This module is presented from the official syllabus scope for Basic Medical Sciences. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: organs
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

3 Understanding Process. Illustrate the structure of ear, process of hearing

3 Understanding Process. Illustrate the structure of ear, process of hearing is an official topic in Module 4 of Basic Medical Sciences. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Process. Illustrate the structure of ear, process of hearing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding process. illustrate the structure of ear, process of hearing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Process. Illustrate the structure of ear, process of hearing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-[illegible] 3 Understanding Process. Illustrate the structure of ear, process of hearing [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Understanding and equilibrium Summarize the functioning of sense of smell,

3 Understanding and equilibrium Summarize the functioning of sense of smell, is an official topic in Module 4 of Basic Medical Sciences. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding and equilibrium Summarize the functioning of sense of smell, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding and equilibrium summarize the functioning of sense of smell, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding and equilibrium Summarize the functioning of sense of smell, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 3 Understanding and equilibrium Summarize the functioning of sense of smell, □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

3 Understanding taste and touch. Special Sense Organs Eye: anatomy of eye - different parts - function of each part, process of vision Ear: anatomy of ear - different parts - function of each part, process of hearing, role of ear in equilibrium of body. Nose & Tongue: Process of sensing of smell and taste - structure of olfactory receptors and taste buds. Skin: Sensory receptors in human skin - Cutaneous receptors – mechanoreceptors - nociceptors and thermoreceptors Text / Reference: T/R Book Title/Author T1 Anatomy and Physiology for Nurses/ Evelyn C Pearce T2 Introduction to Biomedical Instrumentation/ Leslie Cromwell R2 A textbook of Medical Physiology / Artur C Guyton & Hall R3 Handbook of Biomedical Instrumentation / R S Khandpur R4 Applied Physiology / Samson Wright & Cyril A Keele Online resources: Sl.No Website Link 1 www.wikipedia.com 2 www.khanacademy.org 3 https://freevideolectures.com/subject/anatomy-physiology/ 4 https://www.shomusbiology.com/human-physiology.html 5 http://oli.stanford.edu/anatomy-and-physiology/

3 Understanding taste and touch. Special Sense Organs Eye: anatomy of eye - different parts - function of each part, process of vision Ear: anatomy of ear - different parts - function of each part, process of hearing, role of ear in equilibrium of body. Nose & Tongue: Process of sensing of smell and taste - structure of olfactory receptors and taste buds. Skin: Sensory receptors in human skin - Cutaneous receptors – mechanoreceptors - nociceptors and thermoreceptors Text / Reference: T/R Book Title/Author T1 Anatomy and Physiology for Nurses/ Evelyn C Pearce T2 Introduction to Biomedical Instrumentation/ Leslie Cromwell R2 A textbook of Medical Physiology / Artur C Guyton & Hall R3 Handbook of Biomedical Instrumentation / R S Khandpur R4 Applied Physiology / Samson Wright & Cyril A Keele Online resources: Sl.No Website Link 1 www.wikipedia.com 2 www.khanacademy.org 3 https://freevideolectures.com/subject/anatomy-physiology/ 4 https://www.shomusbiology.com/human-physiology.html 5 http://oli.stanford.edu/anatomy-and-physiology/ is an official topic in Module 4 of Basic Medical Sciences. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding taste and touch. Special Sense Organs Eye: anatomy of eye - different parts - function of each part, process of vision Ear: anatomy of ear - different parts - function of each part, process of hearing, role of ear in equilibrium of body. Nose & Tongue: Process of sensing of smell and taste - structure of olfactory receptors and taste buds. Skin: Sensory receptors in human skin - Cutaneous receptors – mechanoreceptors - nociceptors and thermoreceptors Text / Reference: T/R Book Title/Author T1 Anatomy and Physiology for Nurses/ Evelyn C Pearce T2 Introduction to Biomedical Instrumentation/ Leslie Cromwell R2 A textbook of Medical Physiology / Artur C Guyton & Hall R3 Handbook of Biomedical Instrumentation / R S Khandpur R4 Applied Physiology / Samson Wright & Cyril A Keele Online resources: Sl.No Website Link 1 www.wikipedia.com 2 www.khanacademy.org 3 https://freevideolectures.com/subject/anatomy-physiology/ 4 https://www.shomusbiology.com/human-physiology.html 5 http://oli.stanford.edu/anatomy-and-physiology/ using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding taste and touch. special sense organs eye: anatomy of eye - different parts - function of each part, process of vision ear: anatomy of ear - different parts - function of each part, process of hearing, role of ear in equilibrium of body. nose & tongue: process of sensing of smell and taste - structure of olfactory receptors and taste buds. skin: sensory receptors in human skin - cutaneous receptors – mechanoreceptors - nociceptors and thermoreceptors text / reference: t/r book title/author t1 anatomy and physiology for nurses/ evelyn c pearce t2 introduction to biomedical instrumentation/ leslie cromwell r2 a textbook of medical physiology / artur c guyton & hall r3 handbook of biomedical instrumentation / r s khandpur r4 applied physiology / samson wright & cyril a keele online resources: sl.no website link 1 www.wikipedia.com 2 www.khanacademy.org 3 https://freevideolectures.com/subject/anatomy-physiology/ 4 https://www.shomusbiology.com/human-physiology.html 5 http://oli.stanford.edu/anatomy-and-physiology/ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding taste and touch. Special Sense Organs Eye: anatomy of eye - different parts - function of each part, process of vision Ear: anatomy of ear - different parts - function of each part, process of hearing, role of ear in equilibrium of body. Nose & Tongue: Process of

Aspect	What to prepare
	sensing of smell and taste - structure of olfactory receptors and taste buds. Skin: Sensory receptors in human skin - Cutaneous receptors – mechanoreceptors - nociceptors and thermoreceptors Text / Reference: T/R Book Title/Author T1 Anatomy and Physiology for Nurses/ Evelyn C Pearce T2 Introduction to Biomedical Instrumentation/ Leslie Cromwell R2 A textbook of Medical Physiology / Artur C Guyton & Hall R3 Handbook of Biomedical Instrumentation / R S Khandpur R4 Applied Physiology / Samson Wright & Cyril A Keele Online resources: Sl.No Website Link 1 www.wikipedia.com 2 www.khanacademy.org 3 https://freevideolectures.com/subject/anatomy-physiology/ 4 https://www.shomusbiology.com/human-physiology.html 5 http://oli.stanford.edu/anatomy-and-physiology/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 3 Understanding taste and touch. Special Sense Organs Eye: anatomy of eye - different parts - function of each part, process of vision Ear: anatomy of ear - different parts - function of each part, process of hearing, role of ear in equilibrium of body. Nose & Tongue: Process of sensing of smell and taste - structure of olfactory receptors and taste buds. Skin: Sensory receptors in human skin - Cutaneous receptors – mechanoreceptors - nociceptors and thermoreceptors Text / Reference: T/R Book Title/Author T1 Anatomy and Physiology for Nurses/ Evelyn C Pearce T2 Introduction to Biomedical Instrumentation/ Leslie Cromwell R2 A textbook of Medical Physiology / Artur C Guyton & Hall R3 Handbook of Biomedical Instrumentation / R S Khandpur R4 Applied Physiology / Samson Wright & Cyril A Keele Online resources: Sl.No Website Link 1 www.wikipedia.com 2 www.khanacademy.org 3 https://freevideolectures.com/subject/anatomy-physiology/ 4 https://www.shomusbiology.com/human-physiology.html 5 http://oli.stanford.edu/anatomy-and-physiology/ □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
organs
            Describe the structure of the eye and the visual
  M4.01                                                           3
Understanding
            Process.
            Illustrate the structure of ear, process of hearing
  M4.02                                                           3
Understanding
            and equilibrium
            Summarize the functioning of sense of smell,
  M4.03                                                           3
Understanding
            taste and touch.
            Series Test – II                                      1
Contents:
Special Sense Organs
Eye: anatomy of eye - different parts - function of each part, process of vision
Ear: anatomy of ear - different parts - function of each part, process of
hearing, role of ear
in equilibrium of body.
Nose & Tongue: Process of sensing of smell and taste - structure of olfactory
receptors and
taste buds.
Skin: Sensory receptors in human skin - Cutaneous receptors – mechanoreceptors -
nociceptors and thermoreceptors

Text / Reference:
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Define various medical terminologies 2 Remembering Describe the anatomy and physiology of cell and. (3 marks)

Answer: Begin with the standard definition or identity of *Define various medical terminologies 2 Remembering Describe the anatomy and physiology of cell and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding tissues Illustrate the concept of bioelectric potentials. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding tissues Illustrate the concept of bioelectric potentials*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding and demonstrate using Nernst Equation. Extend the concept of bio potential. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding and demonstrate using Nernst Equation. Extend the concept of bio potential*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding transmission in nerves and its measurement. Cell, Tissues and Bio potential Definition of anatomy, physiology, biochemistry, biophysics. Basic anatomical positions and planes - supine, prone, sagittal, coronal and transverse. Cells: structure of human cell, Elementary tissues of the body - types and their functions(listing) - Epithelial, connective, muscle, nerve cells Introduction to bio potential, Origin of bioelectric potential, Resting membrane potential, Action potential, Transmission of impulse through nerve fibers, Recording of membrane potential - microelectrode, monophasic, biphasic. Describe the anatomy and physiology of the cardiovascular and. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding transmission in nerves and its measurement. Cell, Tissues and Bio potential Definition of anatomy, physiology, biochemistry, biophysics. Basic anatomical positions and planes - supine, prone, sagittal, coronal and transverse. Cells: structure of human cell, Elementary tissues of the body - types and their functions(listing) - Epithelial, connective, muscle, nerve cells Introduction to bio potential, Origin of bioelectric potential, Resting membrane potential, Action potential, Transmission of impulse through nerve fibers, Recording of membrane potential - microelectrode, monophasic, biphasic. Describe the anatomy and physiology of the cardiovascular and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 2 Understanding circulation of blood in human body Summarize the anatomy and physiology of. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding circulation of blood in human body Summarize the anatomy and physiology of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding human heart Illustrate the electrical activity of the heart and. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding human heart Illustrate the electrical activity of the heart and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps,

□□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 3 Understanding cardiac cycle Describe the anatomy and physiology of the. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding cardiac cycle Describe the anatomy and physiology of the*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain respiratory system & physics involved in 2 Understanding mechanism of respiration Explain the mechanism of ventilation, lung. (3 marks)

Answer: Begin with the standard definition or identity of *respiratory system & physics involved in 2 Understanding mechanism of respiration Explain the mechanism of ventilation, lung*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Module 3

Define or explain 4 Understanding system and anatomy of brain Summarize the physiology of each part of the. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding system and anatomy of brain Summarize the physiology of each part of the*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding brain Summarize the anatomy and physiology of the. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding brain Summarize the anatomy and physiology of the*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

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Define or explain 2 Understanding urinary system Illustrate the structure of kidneys, nephron and. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding urinary system Illustrate the structure of kidneys, nephron and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding explain the process of urine formation Nervous system and Urinary Systems General organization of the nervous system, Central nervous system - anatomy and functioning of the human brain - cerebrum - cerebellum - brainstem - pons - hypothalamus(temperature regulation) - medulla, Spinal cord Urinary System: anatomy of urinary system - overview of anatomy of the kidney, Structure and function of the nephron, physiology of urine formation Explain the anatomy of the eye and ear and functioning of all special sense. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding explain the process of urine formation Nervous system and Urinary Systems General organization of the nervous system, Central nervous system - anatomy and functioning of the human* *brain - cerebrum - cerebellum - brainstem - pons - hypothalamus(temperature regulation) - medulla, Spinal cord Urinary System: anatomy of urinary system - overview of anatomy of the kidney, Structure and function of the nephron, physiology of urine formation Explain the anatomy of the eye and ear and functioning of all special sense*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain 3 Understanding Process. Illustrate the structure of ear, process of hearing. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding Process. Illustrate the structure of ear, process of hearing*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding and equilibrium Summarize the functioning of sense of smell,. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding and equilibrium Summarize the functioning of sense of smell,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding taste and touch. Special Sense Organs Eye: anatomy of eye - different parts - function of each part, process of vision Ear: anatomy of ear - different parts - function of each part, process of hearing,

role of ear in equilibrium of body. Nose & Tongue: Process of sensing of smell and taste - structure of olfactory receptors and taste buds. Skin: Sensory receptors in human skin - Cutaneous receptors – mechanoreceptors - nociceptors and thermoreceptors Text / Reference: T/R Book Title/Author T1 Anatomy and Physiology for Nurses/ Evelyn C Pearce T2 Introduction to Biomedical Instrumentation/ Leslie Cromwell R2 A textbook of Medical Physiology / Artur C Guyton & Hall R3 Handbook of Biomedical Instrumentation / R S Khandpur R4 Applied Physiology / Samson Wright & Cyril A Keele Online resources: Sl.No Website Link 1 www.wikipedia.com 2 www.khanacademy.org 3 https://freevideolectures.com/subject/anatomy-physiology/ 4 https://www.shomusbiology.com/human-physiology.html 5 http://oli.stanford.edu/anatomy-and-physiology/. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding taste and touch. Special Sense Organs Eye: anatomy of eye - different parts - function of each part, process of vision Ear: anatomy of ear - different parts - function of each part, process of hearing, role of ear in equilibrium of body. Nose & Tongue: Process of sensing of smell and taste - structure of olfactory receptors and taste buds. Skin: Sensory receptors in human skin - Cutaneous receptors – mechanoreceptors - nociceptors and thermoreceptors Text / Reference: T/R Book Title/Author T1 Anatomy and Physiology for Nurses/ Evelyn C Pearce T2 Introduction to Biomedical Instrumentation/ Leslie Cromwell R2 A textbook of Medical Physiology / Artur C Guyton & Hall R3 Handbook of Biomedical Instrumentation / R S Khandpur R4 Applied Physiology / Samson Wright & Cyril A Keele Online resources: Sl.No Website Link 1 www.wikipedia.com 2 www.khanacademy.org 3 https://freevideolectures.com/subject/anatomy-physiology/ 4 https://www.shomusbiology.com/human-physiology.html 5 http://oli.stanford.edu/anatomy-and-physiology/*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Define various medical terminologies 2 Remembering Describe the anatomy and physiology of cell and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **2 Understanding circulation of blood in human body Summarize the anatomy and physiology of**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **4 Understanding system and anatomy of brain Summarize the physiology of each part of the**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **3 Understanding Process. Illustrate the structure of ear, process of hearing**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link www.wikipedia.com www.khanacademy.org
- https://freevideolectures.com/subject/anatomy-physiology/
- https://www.shomusbiology.com/human-physiology.html
- http://oli.stanford.edu/anatomy-and-physiology/

IN-PAGE SEARCH

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